

PCM Paper Figures

Precision Confidence Mapping — action mode network (AMN) targeting. Data, code, and instructions to regenerate Figure 4, Table 1, Figure 5C/D, Supplementary Figure 1, Supplementary Table 1, Supplementary Figure 2, and every other statistic reported in the paper.

CONTENTS

- | | |
|--------------------------------------|---|
| 1. Quick start | 3. Data dictionary |
| 2. Package structure | 4. Figures & tables, one by one |

Quick start

```
cd PCM_paper_figures/code
python3 -m venv .venv && source .venv/bin/activate # optional but recommended
pip install -r requirements.txt
python3 run_all.py
```

That runs all nine scripts in order and writes every figure/table into `outputs/`, mirroring the folder layout described below. Runtime is well under a minute — nothing here needs a cluster. Requires Python ≥ 3.9 ; tested with pandas, numpy, scipy, and matplotlib per `requirements.txt`.

The `data/` folder ships pre-built — you don't need to run `code/prepare_data.py` yourself; it's included only in case you ever need to re-derive the data files.

Package structure

```

PCM_paper_figures/
  instructions.html / instructions.pdf    this document
data/
  combined_metrics/
    combined_metrics_used_in_paper.csv    combined metrics table
  figure5_conditions/
    TM_5min.csv, TM_70min.csv,
    PCM_5min_thr0.99.csv, PCM_70min_thr0.99.csv
  supfig3_nmi/
    standardTM_within_between_stats.csv,
    pcm_thr0.99_matched_duration_pairs.csv
code/
  common.py          shared I/O, stats, and plotting helpers
  prep_data.py       one-time data-prep step (already run; see data/)
  table1_stats.py     Table 1
  supptable1_stats.py Supplementary Table 1
  figure4_heatmaps.py Figure 4, panels A + B
  figure4_pr_recall.py Figure 4, panel C
  supfig1_pr_recall.py Supplementary Figure 1 (heatmap + PR-recall x4 networks)
  supfig2_heatmaps.py Supplementary Figure 2
  figure5_panels_cd.py Figure 5, panels C + D
  other_stats_threshold_gain.py thresholding-gain-by-duration stat
  other_stats_supfig3_nmi.py Supp Fig 3 (whole-map NMI reliability)
  run_all.py          runs everything above, in order
  requirements.txt
outputs/              what run_all.py produces (shipped pre-populated)

```

Data dictionary

data/combined_metrics/combined_metrics_used_in_paper.csv

Long-format table, one row per (subject × threshold × metric × network × exploratory-minutes) combination.

COLUMN	MEANING
Subject	One of PFM3T7T01 – PFM3T7T04 (N=4).
Threshold	PCM confidence threshold, 0.0–0.99. Vertices below threshold are excluded from that map before computing metrics.

Metric	One of <code>PPV_cort</code> , <code>TPR_cort</code> , <code>FPR_cort</code> , <code>TNR_cort</code> , <code>FNR_cort</code> , <code>DC_cort</code> (Dice), <code>MIn_cort</code> (normalized mutual information), plus raw counts <code>TP_cort</code> / <code>FP_cort</code> / <code>FN_cort</code> .
Network	One of the 15 canonical networks: <code>AMN</code> , Aud, DAN, DMN, FP, MTL, PMN, PON, SCAN, SMd, SMI, Sal, Tpole, VAN, Vis.
ExpMin	Length (minutes) of the exploratory data compared against the reference: 5, 10, 15, ..., 70.
RefMin	Length (minutes) of the reference data. Always <code>70</code> (the fixed ground-truth reference used everywhere in the paper).
Value	The metric's value for that row.

`data/figure5_conditions/*.csv`

Four files — `TM_5min.csv` , `TM_70min.csv` , `PCM_5min_thr0.99.csv` , `PCM_70min_thr0.99.csv` — used only for Figure 5C/D. **These have a completely different design from the combined metrics file above:** each row compares one subject's exploratory-half map against one subject's reference-half map, both at the *same* fixed duration (5 or 70 minutes) — this is what lets Figure 5 measure within- vs. between-subject agreement.

COLUMN	MEANING
Subject	Always <code>allsubs</code> — a placeholder, not meaningful here.
SubjIndex_Exp , SubjIndex_Ref	Subject index 1–4 for the exploratory and reference map respectively — <i>not</i> minutes, despite these columns being called <code>ExpMin</code> / <code>RefMin</code> in the original raw files. Renamed here specifically to avoid that trap.
SubjectID_Exp , SubjectID_Ref	Added in this package: the actual <code>PFM3T7T0X</code> ID for each index, so you don't have to cross-reference a separate key.
Threshold , Metric , Network , Value	Same meaning as the combined metrics file.

Diagonal rows (`SubjIndex_Exp == SubjIndex_Ref`) are **within-subject** comparisons; off-diagonal rows are **between-subject** comparisons (averaged over the 3 other subjects per subject, per the manuscript's Methods).

`data/supfig3_nmi/*.csv`

Two files, used only by `other_stats_supfig3_nmi.py`.

pcm_thr0.99_matched_duration_pairs.csv: all 4×4 subject-pair whole-map NMI (`MIn_cort`) comparisons at matched duration (both sides 5 or both sides 70 minutes, PCM confidence maps thresholded at 0.99) — same within/between-subject logic as the Figure 5 condition files above (diagonal = within-subject, off-diagonal = between-subject).

standardTM_within_between_stats.csv: the same within/between comparison for genuine Standard Template Matching, already summarized to mean/SD/paired-t (per-subject-pair values weren't available for this method, only its finished statistics).

Figures & tables, one by one

Table 1

AMN targeting metrics (PPV, FPR, TNR, FNR, TPR) at 5/25/70 minutes, threshold 0 vs. 0.99, paired t-tests across N=4 subjects.

Data	<code>data/combined_metrics/combined_metrics_used_in_paper.csv</code>
Script	<code>code/table1_stats.py</code>
Run	<code>python3 table1_stats.py</code>
Output	<code>outputs/table1/summary_AMN.csv</code> , <code>outputs/table1/stats_AMN.csv</code>

Note: a data-labeling bug in an earlier version of this stats step mislabeled every row's `Network` column `"WB"` whenever `MIn_cort` was included in the same run — cosmetic only, doesn't affect any numeric value; fixed so each row is labeled by its own metric.

Supplementary Table 1

Same design as Table 1, for DMN, FP (called FPN in the paper), SCAN, and Sal (called SAL in the paper) — PPV and TPR only — plus a mean-across-all-15-networks row.

Script	<code>code/supptable1_stats.py</code>
Run	<code>python3 supptable1_stats.py</code>
Output	<code>outputs/supptable1/stats_{DMN,FP,SCAN,Sal}.csv</code> , <code>outputs/supptable1/stats_All15Networks_mean.csv</code>

Figure 4

Panels A+B: AMN PPV heatmap (A) and TNR/FPR/FNR/TPR heatmaps (B), threshold (y) × exploratory data length (x), averaged across subjects; each panel's colorbar auto-scales to its own data range, matching the published figures. **Panel C:** precision (PPV) vs. recall (TPR) for AMN, one line per exploratory data length (5, 25, and 70 minutes), points colored by confidence threshold, with a reference line at precision = 0.6.

Scripts	<code>code/figure4_heatmaps.py</code> (A+B), <code>code/figure4_pr_recall.py</code> (C)
Run	<code>python3 figure4_heatmaps.py && python3 figure4_pr_recall.py</code>
Output	<code>outputs/figure4/Figure4A_PPV_heatmap_AMN.png</code> , <code>Figure4B_*_heatmap_AMN.png</code> (×4), <code>Figure4C_PR_recall_AMN.png</code>

Supplementary Figure 1

Same PPV-heatmap + precision-recall analysis as Figure 4A/C (same 5/25/70-minute lines), repeated for the four clinically-relevant networks: FP (FPN), DMN, SCAN, Sal (Salience).

Script	<code>code/suppfig1_pr_recall.py</code>
Run	<code>python3 suppfig1_pr_recall.py</code>
Output	<code>outputs/suppfig1/SuppFig1_PPV_heatmap_<net>.png</code> , <code>SuppFig1_PR_recall_<net>.png</code> (×4 networks)

The published figure also shows representative thresholded/unthresholded brain-surface maps per network. This package doesn't regenerate those (no surface-rendering tools are included here).

Supplementary Figure 2

Same design as Figure 4, averaged across all 15 canonical networks instead of AMN alone, plus Panel C: whole-map NMI and Dice coefficient heatmaps.

Script	<code>code/suppfig2_heatmaps.py</code>
Run	<code>python3 suppfig2_heatmaps.py</code>
Output	<code>outputs/suppfig2/SuppFig2A_PPV_heatmap_All15Networks.png</code> , <code>SuppFig2B_*.png</code> (x4), <code>SuppFig2C_NMI_heatmap.png</code> , <code>SuppFig2C_Dice_heatmap_All15Networks.png</code>

Figure 5, panels C+D

Panel C: within- vs. between-subject AMN PPV, TM and PCM(thr=0.99), 5 and 70 minutes. **Panel D:** subject-specificity separation (within minus mean-between), same four conditions, paired lines TM→PCM.

Data	<code>data/figure5_conditions/*.csv</code>
Script	<code>code/figure5_panels_cd.py</code>
Run	<code>python3 figure5_panels_cd.py</code>
Output	<code>outputs/figure5/Figure5C_within_between_PPV.png</code> , <code>Figure5D_subject_specificity_separation.png</code> , <code>figure5_stats_ready.csv</code> , <code>figure5_inferential_tests.csv</code>

The "TM" conditions use genuine, independently-run Standard Template Matching data (`ExpMethod-TM` source files) — confirmed to reproduce every reported Figure 5 quantity in the current manuscript exactly, including the Holm-corrected family of 4 tests (Methods: "Holm–Bonferroni correction applied across the four comparisons"). An earlier version of this package used a different source file (PCM confidence maps at threshold=0, standing in for TM) that matched an older draft of the manuscript but not the current one — that's fixed; see the Validation section below for the exact numbers checked.

Other reported statistics

Thresholding-gain-by-duration (main text): is the PPV gain from thresholding (Thr 0.99 – Thr 0) larger at 5 minutes than at 70 minutes? Paired t-test across subjects.

Script	<code>code/other_stats_threshold_gain.py</code>
Output	<code>outputs/other_stats/threshold_gain_by_duration.csv</code>

Supplementary Figure 3 (whole-map NMI within/between-subject reliability, TM vs. PCM@0.99, 5 vs. 70 min):

Script`code/other_stats_supplefig3_nmi.py`**Data**`data/supplefig3_nmi/standardTM_within_between_stats.csv` (genuine Template Matching side) and`data/supplefig3_nmi/pcm_thr0.99_matched_duration_pairs.csv` (PCM side, all 4×4 subject-pair comparisons at matched duration — within-subject vs. between-subject NMI is computed directly from these)**Output**`outputs/other_stats/supplefig3_nmi_within_between.csv`

Unlike every other figure in this package, Supp Fig 3 compares two *independent halves* of data of the *same* duration (Half 1 vs. Half 2, both 5min or both 70min) rather than exploratory-data-vs-fixed-70min-reference — see Supplemental Materials p.6–7. One thing worth knowing about the TM side specifically: Standard Template Matching has no PCM confidence threshold, so it isn't part of the usual threshold sweep — its per-subject raw comparisons weren't available, only its already-computed within/between summary statistics, which are used directly rather than recomputed from scratch.

Built as a reproducibility companion to the PCM manuscript. Original analysis code and raw pipeline outputs live outside this package in the parent project's `analyses/` folder.